Adaptability

One Operation Behind Shape Matching and Analogy, and Why It Is Not Called Intelligence

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Abstract

Two capabilities this programme treats separately — matching a shape against a held codebook, and reasoning by analogy — are one operation at two grains. Both *extract the transform between things already held and apply it where nothing is held.* Naming that operation gives a quantity worth having: *adaptability*, the capacity to acquire a competence that was not supplied. The word is chosen against *intelligence* deliberately. Intelligence is a scalar people rank; adaptability is an operation with a cost, an ordering, and a failure mode, and a system either performs it on a given case or does not.

1 Why not intelligence

Intelligence names a ranking before it names a mechanism. Asking whether a system has more of it than another invites a scalar, and a scalar invites a benchmark, and a benchmark measures what a system already does rather than what it can come to do. That is the wrong quantity for the question here.

Adaptability is narrower and is an operation rather than a rank. A system is adaptable on a case to the degree that, meeting something it does not hold, it ends up holding it — and the interesting part is the mechanism by which the new holding is *derived from* the old rather than supplied. A system given a competence has not adapted. A system that recognises everything it meets has not adapted either; it was already sufficient. *Adaptation requires a residual*, and the size of what must be added to absorb one is the cost.

2 The operation

Definition 2.1 (Adaptation)**.** *Given a held set and an observation not covered by it, an* adaptation *is the addition of the smallest structure under which the observation is covered, derived from what is held.*

Three words carry it. *Smallest*, because an adaptation that adds the observation itself is storage, not adaptation. *Derived*, because an adaptation supplied from outside is instruction. And *covered*, because the test is whether the case is now recognised, not whether a description of it exists.

Remark 2.2 (Shape matching and analogy are one operation). Matching a shape against a codebook and completing an analogy look unrelated because their vocabularies are. They are not.

To decode an observation against held signatures is to ask what transform carries a held one onto it. Where that transform is the identity, the observation is recognised. Where it is a uniform step along a dimension the codebook already has, the observation is a known thing moved — and no new structure is required. Where no such step exists but the residual is *coupled* across the observed dimensions, the transform requires a dimension not currently held, and one must be inferred before the observation can be covered.

To complete an analogy is the same question with the arguments rearranged. Given two things held, the transform between them is extracted; given a third, that transform is applied to reach a fourth which is not held. *Anti-unification — generalising two structures while retaining what distinguishes them — is the general form, and the shape ladder is that form restricted*: at one rung the difference is a single step along a known dimension, at the next it is a dimension.

So the two are not analogous to each other. They are the same operation, and the second is what the first looks like when the objects are rules rather than observations.

3 The ladder, and why its order is not arbitrary

An adaptation must be the smallest available, and that gives an ordering rather than a preference:

1. **Recognise.** The observation is covered as held. Nothing is added, and no adaptation has occurred — which is the correct outcome for most observations.

2. **Transform in the held basis.** One uniform step along a dimension already carried. The observation is a known thing displaced; the codebook does not grow.

3. **Infer a dimension.** The residual is coupled across observed dimensions in a way no step in the current basis produces, so a dimension is minted and the transform is expressed in it.

4. **Mint.** No transform reaches the observation, so it is added as new.

Only rungs three and four are adaptations, and only the third is interesting: it is where the system's own vocabulary changes. Rung four is the honest floor — something genuinely new — and a system that reaches it too readily is not adapting but accumulating.

Remark 3.1 (Why the order must be Occam and not preference). Attempting mint first would always succeed, so a system that did would never transform, never infer, and would grow linearly in what it meets. Attempting transform first is what makes minting *informative*: a mint that survives the earlier rungs is evidence that nothing held reaches the case, which is a claim about the held set and not only about the observation.

That is also why the ordering is a commitment and not an implementation detail. *A system that mints where a transform would have sufficed has recorded a coincidence as a category*, and one that transforms where a mint was required has folded a distinction it needed. Both are failures of adaptation, in opposite directions, and the ladder is what makes them separable.

4 What adaptability predicts

The account is worth having only if it says what an adaptable system does that an unadaptable one does not, on a case neither has seen.

One mint, then recognition. Meeting a genuinely novel case repeatedly, an adaptable system should mint *exactly once* and recognise every later instance — not mint per encounter, which is storage, and not fail to mint, which is refusal. The signature is a step followed by silence.

Zero mints under transformation. Meeting a known thing transformed along a held dimension, it should add nothing at all. A system that mints there is treating the dimension it already has as absent.

Coupling forces dimensions. A residual that is coupled across observed dimensions should produce a dimension, not a member. *The distinguishing case is the one whose difference cannot be written in the current basis at all*, and a system without rung three cannot tell it apart from novelty — it mints where it should have generalised, and the codebook grows with the world instead of with its structure.

And competence should transfer without being carried. If the operation is real, structure built for one domain should be usable unmodified in another whose existence was not anticipated when the structure was made. That is the strongest available test, because it removes the author from the loop, and it is the one the assessment companion reports against.

5 What this does not claim

Adaptability is not a quantity a system has more or less of in general. It is defined *per case* and against *a held set*, so the same system is adaptable on one observation and not on another, and moving the held set moves the verdict. There is no frame-free adaptability, for the same reason this programme's base claim gives for there being no frame-free anything.

Nor does adaptation imply improvement. An adaptation is an addition that covers a residual; whether covering it was worth the cost is a separate question with a separate answer, and a system that adapts readily to noise is worse than one that refuses.

6 Capability has more than one source

There is a temptation, on meeting the account above, to ask whether adaptation is *the* mechanism by which a system comes to do something. The companion programme contains a measurement that answers that question, and the answer is that the question was malformed.

Where verification is cheap and exact, search does the work and holding does not pay. In program synthesis the machinery above is present — a held codebook proposes, an enumerator searches, an execution verdict decides — and stripping the learned content out changed nothing whatever. The two arms did not merely score alike; they solved the *identical set of tasks*, with the difference measured at zero repeatedly. The capability sat on the enumerator and the verdict.

Where verification is expensive or unavailable, holding is the only route. In one-shot character recognition the same programme measured the opposite: composing held parts, rather than treating a character as a bag of them, is worth twenty-one points and is the one effect in that arc that survived a powered re-measurement with intervals separated. Nothing there can be checked cheaply — a candidate reading of an unfamiliar character has no oracle — so search has nothing to search against, and what is held is what decides.

Remark 6.1 (The two are not rivals, and the regime is the question)*.* Read as a contest those measurements conflict. Read as a regime split they do not, and the split is the content: *enumeration substitutes for a codebook exactly where candidates can be checked, and cannot substitute for one where they cannot.* A system with an exact cheap oracle need hold nothing, because it can afford to propose everything; a system without one must hold structure, because proposing everything buys it no way to tell which proposal was right.

That is the same shape this programme keeps arriving at rather than a new one. Its measurement discipline already runs on *bins propose and continuous decides*; its account of the substrate already separates what acquires from what executes; the ladder of Section 3 is itself four sources with different regimes rather than one mechanism with exceptions. *Asking which single source carries capability is the error the rest of the account is arranged to avoid*, and this section exists because the first version of it made that error — reporting a regime as a limit.

What follows is a prediction rather than a reconciliation, and it is the useful part. *Adaptation should pay in inverse proportion to the cheapness of verification.* A domain with an exact, cheap oracle in which held structure nevertheless contributes measurably would be evidence against this reading; so would a domain with no oracle at all in which holding contributes nothing.

And one uncomfortable consequence should be stated rather than left implicit. In the synthesis case the ladder was available and the domain did not need it. That is not a failure — nothing broke, and no prediction of this account was falsified — but it means *the mechanism can be present and idle with nothing indicating so.* A system cannot tell from the inside whether it is adapting or merely searching in a regime where searching suffices, and neither can an observer without the ablation that was run here.

7 What the held set costs, and who paid for it

Definition 2.1 says an adaptation is *derived* from what is held. That word has a consequence the account cannot avoid and should not want to: *adaptability is bounded by the priors.* Nothing can be derived that has no path from the held set, so a system's reach is set by what it already carries, and a system carrying nothing is not maximally general but incapable.

Priors are expensive, and the expense is usually invisible because somebody else paid it. A human takes the better part of a year to walk and longer to do it well. Visual competence is acquired rather than shipped, which is most visible in species born least mature. By the time a person is capable of the tasks we use to demonstrate capability, the overwhelming majority of what makes them capable was assembled over years, and none of it is counted when the demonstration is scored.

That is the informal form of a formal result. Wolpert and Macready (1997) establish that no search or learning procedure outperforms another averaged over all problems; performance on the problems actually faced is therefore bought, and what buys it is a prior matched to them. *The more general the machinery, the more the mapping carries* — generality does not remove the need for a matched prior, it relocates the whole of the burden onto the match. There is no free lunch, and an account of adaptability that implied otherwise would be claiming one.

Proposition 7.1 (There is no prior-free measurement)**.** *A capability cannot be measured against a prior-free baseline, because no such baseline is a member of the space being measured.*

Proof. Measuring a capability *relative to priors* requires a zero on the priors axis: a frame holding nothing, against which holdings can be counted. A frame holding nothing stands in no relation and draws no distinction, and by the companion's first proposition that is not a frame with nothing in it but no frame at all — the set of frames with no particularity has no members, the only object answering to the description being the empty set, which is not a frame and does nothing.

So the zero of the priors axis lies outside the space. An axis so constituted supports comparison between members and does not support measurement of a member against its zero, and a quantity defined by that measurement is the frame-free denominator Axiom −1 refuses. □

The consequence is not that such measurement is difficult. It is that the thing being asked for does not exist to be approximated, which makes it a category error rather than an unsolved problem — and one established by the account's own earlier results rather than assumed here. Most reported measures of intelligence are flawed for that reason and not for want of care: the control they would need is not merely absent, it is not constructible.

Remark 7.2 (What survives, and where this account differs from Chollet)*.* Comparison survives, and less of it survives than the previous sentence of this paragraph once claimed. What a test yields is not a property of a system and not quite a relation between two systems either. It is a property of a four-place arrangement: *a particular graph, in a particular state, inside a*

particular supergraph state, on a particular test. Move any of the four and the result is a different measurement rather than a noisier estimate of the same one.

Inferences from it are licensed only relatively. Within the arrangement, that this graph out-performed that one is established. Outside it nothing follows — not that the ordering holds in another state, not that it holds under a different container, not that the margin would survive a second test — without a separate argument that the moved coordinate does not bear, and such an argument is rarely available and never free.

That is why this programme's own measurement records read as they do. Its enumerated failure baseline names the commit, the compiled engine by hash and path, the wall-clock, the load average at start, and which optional dependencies were present and which absent; its psychological runs name the machine and whether the box was shared. *Those are not hygiene. They are the coordinates*, and a figure reported without them does not underspecify a measurement — it fails to name one.

The cost is real and is the right cost. Comparing across two such records requires establishing that the coordinates match or that the difference does not bear, which is laborious and frequently impossible, and the alternative is a number that reads as portable and is not.

Chollet (2019) takes priors seriously in the same direction and does the harder half of the work: intelligence as the efficiency of skill acquisition *relative to priors, experience and generalisation difficulty*, over a corpus built explicitly on Core Knowledge (Spelke and Kinzler, 2006). This account owes that framing the observation and differs in two places. Priors here are not a term the measure is taken relative to — *they are the held set the operation runs on*, and the ladder is what converts them. And the relativisation is not a methodological refinement that a better experiment could tighten: by Proposition 7.1 it is the only form such a measure can take.

Machinery and mapping cannot be separated from outside either, and for a different reason. Two systems differing in both — one with a stronger operation on a poorer prior, one with the reverse — can agree on every item of any finite test. The decomposition is therefore underdetermined by measurement rather than hidden from it, and no amount of testing at a fixed grain settles it. What refinement buys is approach: a model of the split can be made arbitrarily fine and its residual driven toward zero without ever reaching it, which is this programme's ordinary intensive limit rather than a special defeat. *You can approach one with unbounded analysis. You cannot arrive, and no finite experiment stands in for arriving.*

That yields the prediction the previous section left standing, now in its proper form. A person lacking familiarity with a task set's conventions should score near its floor — not because they are less capable, but because *the score was never a property of the person.* Human baselines for that corpus exist (LeGris et al., 2025); whether they stratify by familiarity in a way that tests this has not been checked here, so it is recorded as untested.

8 What an oracle costs, and how often

The regime argument above is about capability and says nothing about price. Price runs the other way, and where the oracle is expensive it dominates.

An oracle consulted at inference is paid per use, without end. An oracle consulted once, whose output is held, is paid once. That is the whole of the asymmetry, and it is unaffected by how good the oracle is. A system that answers by calling something expensive has bought an answer; a system that calls the same thing once, holds what comes back as structure, and answers from the structure thereafter has bought a capability. The second transaction terminates.

Two consequences follow and the second matters more.

The first is arithmetic. Where an oracle is cheap per call, calling is correct and holding is premature; where it is expensive, the crossover arrives quickly and holding wins on cost even in the regime where searching would win on capability. *So the two arguments can point in opposite*

directions, and which dominates is a fact about the price of the oracle rather than about the account.

The second is that the two purchases differ in kind. What is held is inspectable, mutable, and can be repaired by the system holding it; what is called is none of those, permanently. A capability bought per call sits outside whatever boundary the system's own self-correction runs to, and stays outside no matter how many times it is bought. *Paying once and holding the result moves a capability inside that boundary; paying repeatedly guarantees it never enters.*

Remark 8.1 (The cost of buying once, stated plainly)*.* This is not free and the bill is easy to overlook. *What is held is what was bought on the day it was bought.* A per-call oracle improves when the oracle improves, silently and without the caller doing anything; held structure does not, and a system that bought its competence once is carrying a snapshot. Against that, held structure can be audited and corrected by the system carrying it, which a call cannot be — so the exchange is improvement-for-free against correctability, and neither term is obviously larger.

The companion programme is an instance rather than a demonstration of this. Roughly half its implementation was performed by a language model rather than by its own rule-writing, and the running system makes no such calls at all — a property enforced by a switch and a gate rather than asserted. So the arrangement described here exists and runs. *What does not exist is a control*: nobody has built the same substrate without that development-time help, so the size of what was bought is not known, and the companion on assessment says so at greater length.

9 What follows about persons, and what it costs

One consequence has been left implicit across this programme and is better stated. *It requires no new posit and no new machinery*, which is the only reason it belongs in a paper rather than in a manifesto.

Three things are already held. Whatever exists stands in relation and coheres, so a person is relata and relations like anything else. Nothing is privileged by characterisation, so no exemption for persons is available that some frame does not supply. And the substrate paper states, unhedged, that where every measurement that can be made of a thing and every operation that can be performed on it are captured by treating it as a typed graph, *no difference remains for the question "but is it really" to name* — functional identity there is not a weaker sense of identity but the whole of what identity could mean without a view from nowhere.

Conjoin them. *A graph functionally identical to a person is a person, in the only sense of "is" this account admits*, and building one is therefore not blocked by anything in the account. That is the claim, and it is arrived at by adding nothing.

Remark 9.1 (Neither new nor magic, and the difference is a price)*.* The claim is old. Turing (1950) put indistinguishability at the centre of the question and declined the metaphysics, and functionalism (Putnam, 1967) is the position that a functional duplicate is a mind, with its own developed critical literature (Block and Fodor, 1972). *Nothing here disputes or improves on any of that*, and a reader who takes this section as a novelty claim has misread it.

What this account adds is a *bill.* Functionalism says a duplicate would be a mind and is silent on what a duplicate costs; the question of construction is left as an engineering detail below the philosophy. Section 7 says what the detail contains: the duplicate's competence is its held set, held sets are acquired rather than shipped, and acquisition is the slow expensive thing that takes a human a year to walk and a decade to reach the priors a task set quietly assumes. *The barrier is not principle and it is not mystery. It is the priors, and they are enormous.*

So the honest form of the claim is narrow and is worth stating in one line. *There is no barrier of principle available from within this account, the work is nameable, and naming it does not make it small.* That is a research programme rather than a prediction, and specifically it is not a claim that this substrate is close to it — the companion reports that the faculty this

programme has for acquiring what it was not given is not yet resident where the system could repair it.

Two limits belong with it rather than after it. *Indistinguishable is a relation and not a property*, by the same argument as Proposition 7.1: indistinguishable to some frame, at the frontier where distinguishing currently stops, and that frontier is defined to move. And nothing above asserts that the account is *true* — it says what follows if it is, which is what a consequence is, and the falsifiers elsewhere in this corpus are where the antecedent is put at risk.

10 Prior positions

None of the machinery here is new and it should not be dressed as though it were. Four literatures own it, and what this account adds is an arrangement rather than an operation.

The relativity of measurement, which is old and is not claimed here. Everything Section 7 says about scores has been said, in some cases seventy years ago, and it should be read as inherited rather than offered.

Generalizability theory decomposes a measurement into facets — the person, the item, the occasion, the observer — and holds that a score is a *sample from a universe of admissible observations* rather than a property of whoever was measured (Cronbach et al., 1972). That is the four-place arrangement, in a literature that had it before this programme existed. Cronbach and Meehl (1955) had already established that a test measures a construct only under an argument and never by fiat. Quine (1951) gives the inference-licensing half: no hypothesis meets the evidence alone, so what a result licenses depends on auxiliaries that travel with it. And Newell (1973) made the corresponding objection to task-by-task measurement in this field specifically — that a pile of task results does not accumulate into an account of the mechanism.

One thing differs, and it is small and different in kind. Those are cautions, and cautions are discharged by better practice: estimate more facets, state the auxiliaries, argue the construct. What Proposition 7.1 adds is that the prior-free case is not a facet that has been estimated badly — *the origin is not in the space.* Generalizability theory supplies the coordinates and assumes a universe of admissible observations within which they vary; the proposition denies there is a zero for one of them to be measured from.

The practical force of that difference is the whole of it. A caution tells a careful practitioner to do more work; *a consequence tells them the work does not exist to be done*, and stops a research programme from being organised around obtaining a control that cannot be built. Nothing else here is new, and the derivation is the contribution rather than the observation it derives.

Anti-unification. The operation is Plotkin (1970)'s least general generalisation: given two structures, produce the most specific structure of which both are instances. That is exactly rung two and rung three read backwards. *What differs is what is kept.* The classical formulation returns the generalisation and discards the substitutions that recover the inputs from it; here the substitutions are the point, because *the witness is the transform* and applying it elsewhere is the whole operation. An lgg that throws away its witness cannot adapt anything.

Structure-mapping. Gentner (1983) established that analogy runs on relational structure rather than surface features, and that systematic relational systems are preferred to isolated matches — both of which this account assumes rather than argues. *What differs is the purpose.* Structure-mapping is a theory of comparison: what makes two things analogous and how a person aligns them. This is a theory of acquisition, in which the comparison is instrumental — the alignment exists to yield a transform that mints a competence the system did not have. Gentner's theory answers what an analogy *is*; the question here is what one is *for*.

Analogical proportion. The four-term form, a is to b as c is to d, has a developed logical treatment (Prade and Richard, 2014), and parallelogram completion is that form with d unknown. Taken directly and without modification.

Embedding arithmetic, and the reason its critique does not transfer. The word *parallelogram*

comes from vector offsets in learned embedding spaces (Mikolov et al., 2013), and that method's evaluation has been substantially criticised: Levy and Goldberg (2014) showed its behaviour is largely accounted for by similarity terms rather than by the offset, and Linzen (2016) showed the standard benchmark is confounded by nearest-neighbour baselines that need no analogy at all.

Those criticisms are correct and they do not apply here, for a reason that is a real difference and not a defence. The offset method is metric: it returns the nearest point to a computed location, so it always returns something, and the question is only whether the something is right. The operation here is symbolic and exact — anti-unification either yields a witness that is well-defined and lossless, or it fails and yields nothing. *There is no nearest neighbour to fall back on and no threshold to tune*, which removes the failure mode the critique identifies and replaces it with a different one: refusing cases a metric method would have answered. That is the cost of the exchange and it is the honest form of it.

11 Falsifiability commitments

1. [SHAPE] **Shape matching and analogy are one operation.** Fails if a case of one cannot be expressed as a case of the other — specifically, if some analogical completion has no formulation as a transform between held signatures, or some shape adaptation has no formulation as an anti-unification with its witness retained.

2. [SHAPE] **The ladder's order is forced, not chosen.** Fails if reordering the rungs leaves the outcomes unchanged on cases where the rungs disagree, which would make the order a preference.

3. [DETAIL] **One mint, then recognition.** Fails on a novel case that yields repeated minting under an unchanged codebook, or none at all.

4. [DETAIL] **Coupling forces a dimension.** Fails if a residual coupled across observed dimensions is absorbed by a member rather than a dimension, without loss.

5. [SHAPE] **Adaptability is bounded by the priors.** Fails on a demonstration of a competence derived from a held set containing no path to it — which would make adaptation a source of capability rather than a conversion of one, and would be a free lunch.

6. [SHAPE] **No prior-free baseline is constructible.** Fails on the exhibition of a frame holding nothing that nevertheless draws a distinction — which would put the zero of the priors axis inside the space and make prior-free measurement well-formed after all.

7. [DETAIL] **A prior-assuming benchmark measures a relation, not a property.** Fails if performance on such a task set is shown to be independent of familiarity with its conventions, in a population matched on everything else.

8. [DETAIL] **An oracle bought once is paid once.** Fails if held structure derived from an oracle requires continued access to that oracle to keep working — if the purchase does not in fact terminate, the distinction drawn here is between two ways of paying continuously and is not worth drawing.

9. [SHAPE] **Capability has plural sources, selected by regime.** Fails if a domain is exhibited in which candidates are cheaply and exactly checkable and held structure nevertheless contributes measurably, or in which no oracle exists and holding contributes nothing — either would break the inverse relation this account asserts between the value of holding and the cheapness of verification.

10. [SHAPE] **Adaptability is per case, not a rank.** Fails if a single ordering of systems by adaptability can be constructed that survives changing the held set.

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